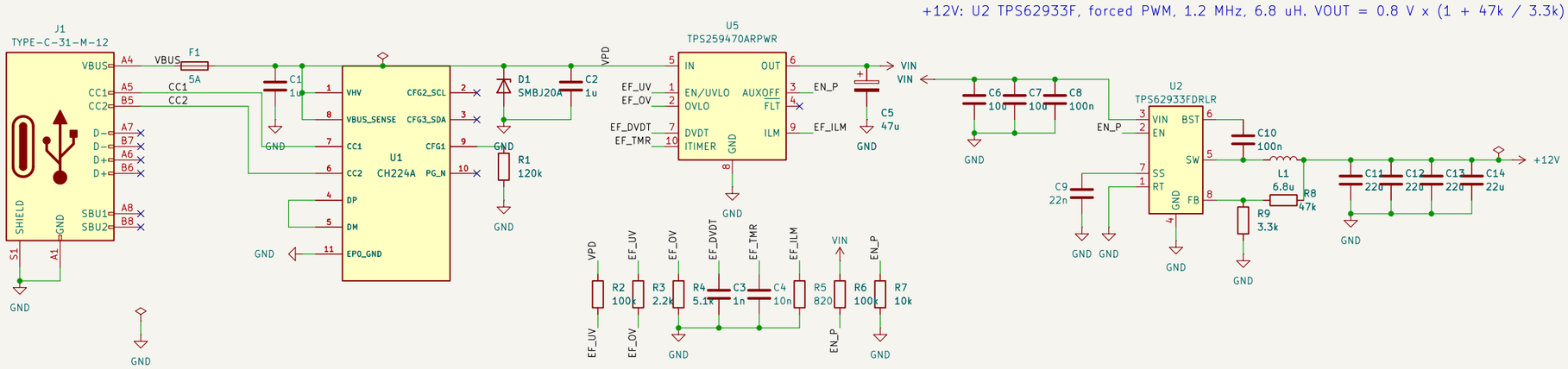
USB-C PD 20 V INPUT: CH224A (CFG1 120k = 20 V), 5 A fuse, SMBJ20A TVS. VBUS sees only C1 + C2 (2 uF) at attach. Use a 20 V PD charger rated 45 W or more; 60 W for the full budget

U5 TPS259470A eFuse: on 17.6–25.3 V only (R2/R3/R4), VIN ramps at 2 V/ms (C3), limit 4.1 A (R5), auto-retry. AUXOFF holds EN_P low until VIN has ramped, then U2/U3 start

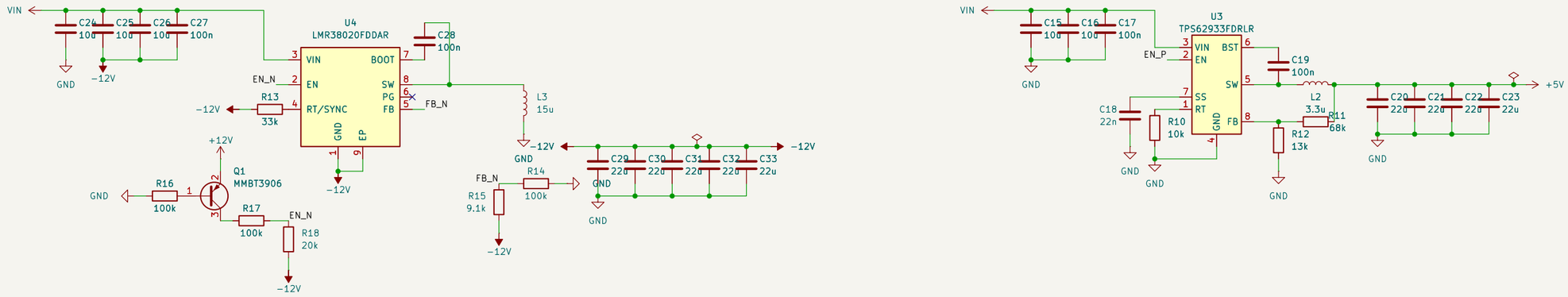
+12V: U2 TPS62933F, forced PWM, 1.2 MHz, 6.8 uH. VOUT = 0.8 V x (1 + 47k / 3.3k) = 12.19 V



−12V: U4 LMR38020F inverting buck-boost, forced PWM, ~790 kHz (RT 33k), 15 uH. IC ground pins = −12V. |VOUT| = 1.0 V x (1 + 100k / 9.1k) = 11.99 V

Q1: −12V is enabled from +12V (starts at +12V ~ 8 V, stops when +12V goes): the −12V rail never runs alone. Internal compensation

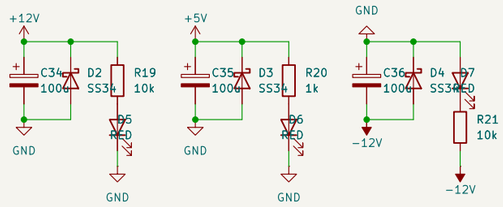
+5V: U3 TPS62933F, forced PWM, ~2.0 MHz (RT 10k), 3.3 uH. VOUT = 0.8 V x (1 + 68k / 13k) = 4.98 V



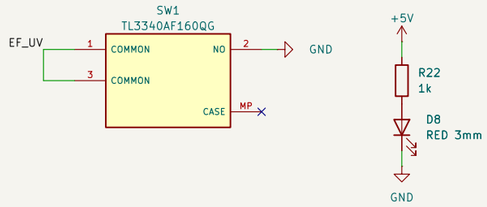
OUTPUT: 2 x 16-pin Eurorack box header in parallel (Doepfer pinout, pin 1 = −12V)



Bulk caps, rail clamps (SS34) and rail LEDs at the headers



FRONT PANEL (as usb-power): SW1 RESET pulls U5 EN/UVLO low: all rails off, release = restart. D8 red 3 mm +5V power LED (hand-soldered)



Budget (docs/design.md): +12V 2.5 A, +5V 2.5 A, −12V 1.0 A per rail; all rails together <= ~39 W out on a 45 W charger (~52 W on 60 W)

Design notes and budget: docs/design.md
S3: PD 20 V -> eFuse -> forced-PWM switchers; no LDOs; RESET + panel LED

Big Modular

Sheet: /
File: usb-power-45w.kicad_sch

Title: Big Modular — usb-power-45w

Size: A3	Date:	**Rev: S3**
KiCad E.D.A. 10.0.6		Id: 1/1